Phase 2 | Requirement Analysis

Technology Stack

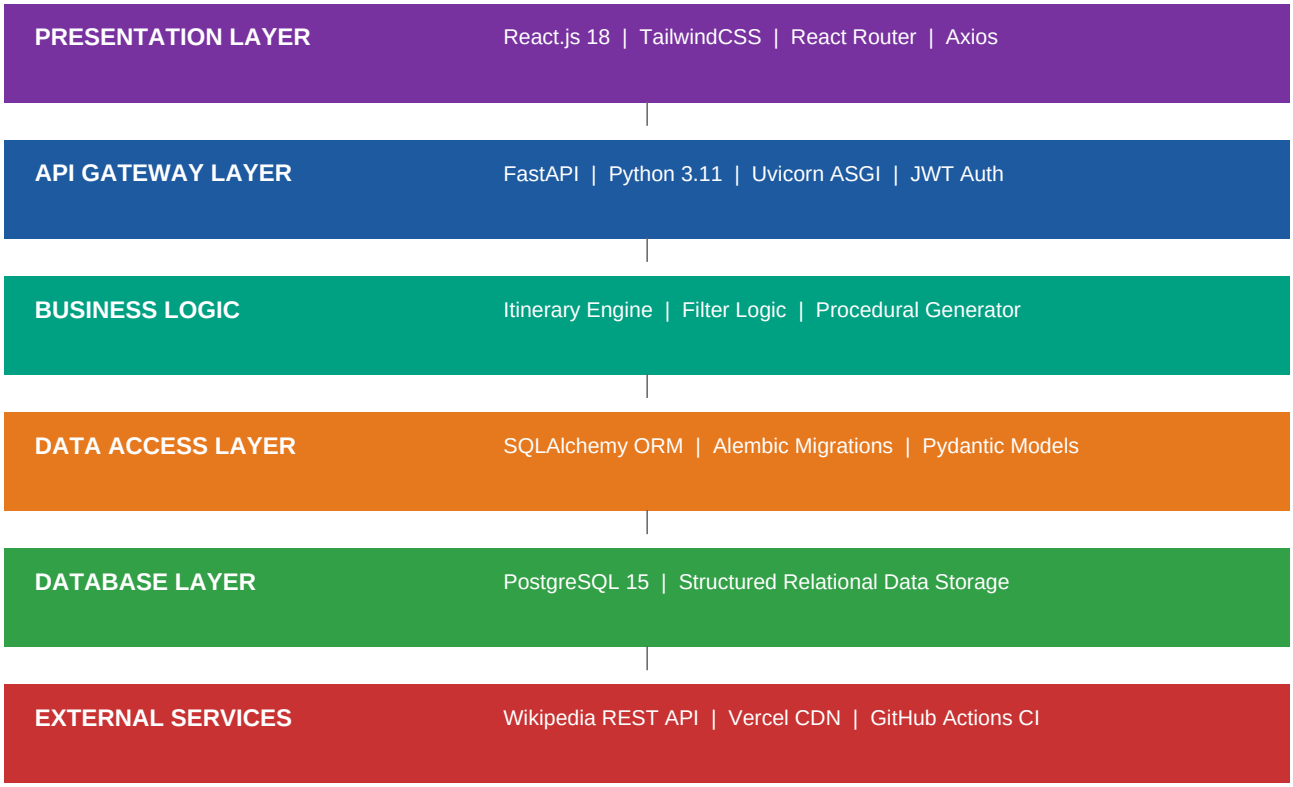
AI-Powered Travel Planner | Smart Situational Itinerary Engine

Live URL:	https://travel-planner-iota-plum.vercel.
GitHub:	github.com/itskian2007/travel-planner
Tech:	React + FastAPI + PostgreSQL + Vercel
Team:	itskian2007

1. Technology Selection Philosophy

Technologies were selected based on four criteria: (1) Industry adoption & community support, (2) Development speed, (3) Scalability ceiling, and (4) Free/open-source availability. The final stack provides a production-grade, fully deployable AI-ML web application.

2. System Architecture Layers Diagram



3. Technology Deep Dive

React.js 18

- Component-based UI architecture
- Virtual DOM for fast re-renders
- Hooks: useState, useEffect, useContext
- React Router v6 for SPA navigation
- Deployed on Vercel with CDN distribution

- Async Python ASGI framework

- Auto-generates OpenAPI/Swagger docs

- Built-in data validation via Pydantic

- JWT authentication middleware

- 60k+ req/sec throughput capability

PostgreSQL 15

- ACID compliant relational database
- JSONB columns for itinerary data
- Full-text search for destination queries
- Connection pooling via SQLAlchemy
- Row-level security for user data

Wikipedia API

- REST API for free image sourcing
- Fetches verified landmark photographs
- No API key required (rate limited)
- Used for all 304 destination images
- Fallback logic for missing images

4. Technology Comparison & Selection Rationale

Layer	CHOSEN	Alternative 1	Alternative 2	Decision Rationale
Frontend	**React.js**	Vue.js	Angular	React chosen for ecosystem size & Vercel integration
Backend	**FastAPI**	Django	Express.js	FastAPI chosen for async performance & auto-docs
Database	**PostgreSQL**	MongoDB	MySQL	PostgreSQL for ACID compliance & JSONB support
Hosting	**Vercel**	Heroku	AWS	Vercel chosen for free tier & zero-config deployment
Auth	**JWT**	Sessions	OAuth2	JWT for stateless, scalable API authentication
Images	**Wikipedia API**	Unsplash	S3	Wikipedia API for free, verified landmark images